

TrES-3b

R-2547

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_110423 · created 2026-08-26T00:37:54+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2455674.96269 +/- 0.00087 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.151 +/- 0.016
Transit depth (Rp/Rs)^2	2.27 +/- 0.48 [%]
Orbital Inclination (inc)	84.4 +/- 0.75
Airmass coefficient 1 (a1)	1.553 +/- 0.013
Airmass coefficient 2 (a2)	-0.42 +/- 0.14
Scatter in the residuals of the lightcurve fit is	0.83 %
Best Comparison Star	None
Optimal Aperture	3.74
Optimal Annulus	12.92
Transit Duration (day)	0.0709 +/- 0.0039

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.151 ± 0.016 vs 0.16309 (deviation 0.76σ)
Duration fit vs published	0.0709 d vs 0.0567 d (deviation 3.65σ)
Mid-time O–C	-1.8 min
Depth SNR	9.4
Residual scatter	0.83 %

Light curve

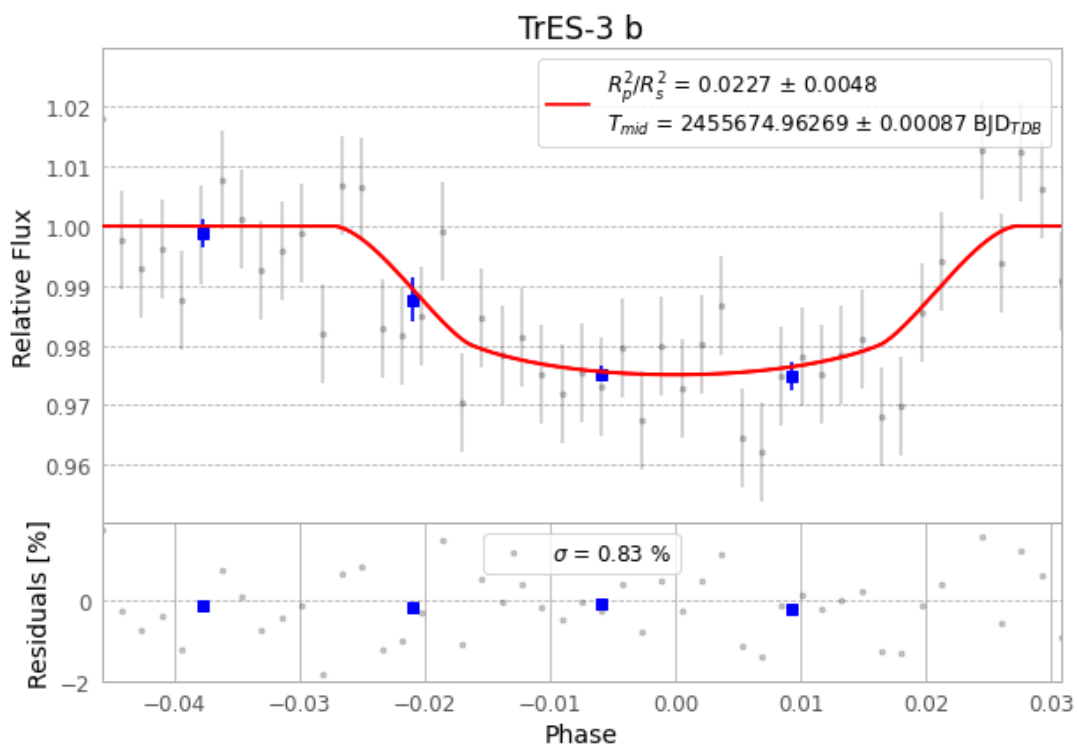


Figure 1 — FinalLightCurve_TrES-3 b_2011-04-23.png (SHA-256 4e8d94da02f711c7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [225, 169], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 748ceb25fda6b11b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

49 FITS frames (32 MB), TRES-3110423093613.FITS → TRES-3110423120013.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-110423.log (1de22e74c190...), FinalLightCurve_TrES-3 b_2011-04-23.png (4e8d94da02f7...), FinalLightCurve_TrES-3 b_2011-04-23.csv (4a2070f66a68...)

Compute resources

Wall time 226.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-26 00:37Z.